

A
Project Report
on

JOB PORTAL

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Computer Science Engineering 2025-2026

Guided By
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Submitted to



MGM's College of Engineering, Nanded

Under

Dr. Babasaheb Ambedkar Technological University, Lonere.

Certificate

This is to certify that the report entitle

JOB Portal

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in satisfactory manner as a partial fulfillment of

Third year in Information Technology To

MGM's College of Engineering, Nanded

Under

Dr. Babasaheb Ambedkar Technological University, Lonere

has been carried out under my guidance,

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With Deep Reverence,

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ABSTARCT

Nowadays, very few people move from office to office depositing their resumes. The possibility of looking for jobs online has made this process seamless and less time-consuming. Indeed, alone has over 200 million resumes. This means that it is very profitable and convenient to create a job search website that helps employers fill vacant positions. Due to the COVID-19 pandemic, many people have lost jobs and were compelled to discover alternative work options. So, people started using job search websites (like Indeed, Naukri and LinkedIn, etc.) actively to look for employment possibilities. And these sites are really powerful tools.

It is a platform where employers list their job offers and search for relevant candidates. On the flip side, applicants post their resumes and look for the desired vacancies. Also, the job website stands for a worldwide search engine. It aggregates job postings from many other employment services and places them on the platform. To complete this project, we'll be using technologies/tools such as HTML, CSS, JavaScript, And PHP (for backend) And for database management system MySQL.

INTRODUCTION

Technology is constantly changing. Society as we know it depends on this fact. That which we take for granted today would have been the stuff of science fiction as little as fifty years ago. In fifty years' time, we will doubtless be excited, perturbed and baffled by yet more new Developments. In this years of the twenty first century, it is computers and the internet that have captured the public imagination, and found their way into not just the working environments, but increasingly into the domestic spaces.

In this modern society, if we are not capable to cope up with these changes than we are not going to stand or survive anywhere in this technical world. Today there is no place for errors, so as to make a system more effective and efficient we need such technology where error prone chances must be least.

In the scenario of the assignment we are required to developed a web-based application on job Portal Management System. In this time of recession where everyone, is either experienced or fresher, is in search for a job. This job portal can prove to be very helpful since it allows users of different profile to upload their CVs, search job on the basis of their qualification. Every user can access through user id and apply multiple jobs at a time.

Currently, we are working on a manual system where data is stored in the form of registers viewing available jobs, or applying for the job at the agency can be done for which job seekers has to go the agency and check and the available jobs at the agency. Job seeker check the list of jobs available and apply the job. Then the agency will show available jobs foe the job seeker for his qualification and then updates the jobs database. The developed job Portal management system is web-based which requires Employee Registration & profiles, job Search, Employer Registration & Profiles, and Subscription Option for both Employee and Employer etc. Employer can add Own profile and post jobs and Jobs Seeker can Search Jobs based on Geographical Area (Country, State and City), Qualification, and Company wise or based on Experience and Expertise wise.

The cost of searching/posting jobs will be much less compared to the traditional way of advertising. Job search portal stands as an effective means for Employers to outline the job vacancies, responsibilities, and qualifications to attract jobseekers. Using the portal jobseekers can extensively search for jobs in companies, organizations, and regions they may otherwise have not learnt. In addition, candidates/Employers can write a review about an organization, which might help them to change the way things are done.

The purpose of the designing the online job portal is to give the job seekers a platform for finding a right and a satisfactory job according to their qualification. It also connects the job seekers with the major agencies.

It also provides Jobs portal for Job Seekers to submit their CV and apply for job posting and Employer can select best Employees from Available CV based on their payment option selection. This is basically a Job portal where job Seeker applies for jobs and employer post jobs and select prospective applicant.

Job portal is prepared for provide all categories of job and help to get various type of job. The main purpose of job portal is to provide the facility to job seekers for getting the quick job. So, it enables applicants to search for jobs in a convenient manner and to enable employers to find suitable candidates.

1.1 Objective

The objective of the application is to develop a system using which job applicants and recruiters can communicate with each other. An Online Job Portal is an application where the job seekers can register themselves at the website and search jobs which are suitable for them where as the employers register with the website and put up jobs which are vacant at their company.

The Online Job Portal System is a package to be used by agencies to improve the efficiency of business. The Online Job Portal System to be developed benefits greatly the members. The system provides jobs catalogue and information to members and helps them decide on the jobs to apply. The Admin and employers can keep the jobs catalogue updated all the time so that the Job seekers get the updated information all the time.

The online job Portal System that is to be developed provides the members with jobs information, online applying for jobs and many other facilities. This system provides service to the job applicants to search for working opportunities. Job Portal will allow job provider to establish one to one relationships with candidates. This Portal will primarily focus on the posting and management of job vacancies. This system is designed such that ultimately all vacancies will be posted online and would offer employers the facilities to post their vacancies online. It helps to review and manage the resulting applications efficiently through the web. Employer can also find the resume according to key skill in very less amount of time.1.3 Scope

As of Indian market, there is ample opportunities for the job portal sites, as more and more number of educated and skilled young people are coming out each and every year. Also, as the growth rate of India is zooming to be at a healthy rate over 7%, so it is boom time for corporate also. So, more and more number of lucrative careers will be available for the job seekers. So, it is now the right period for the job portal sites to think out of the box, and to make most of the opportunities available.

The Scope for the system can be as follow

- Maintain job seeker and Employer records.
- Maintain uploaded resumes.
- Provide customized job posting.
- Maintain job Posting details and generate various report.

1.4 Features of the Job Searching Portal

- Product and Component based
- Creating & Changing Issues at ease
- Query Issue List to any depth
- Reporting & Charting in more comprehensive way
- User Accounts to control the access and maintain security
- Simple Status & Resolutions
- Multi-level Priorities & Severities.
- Targets & Milestones for guiding the programmers
- Attachments & Additional Comments for more information
- Robust database back-end
- Various level of reports available with a lot of filter criteria's
- It contain better storage capacity.
- Accuracy in work.
- Easy & fast retrieval of information.
- Well designed reports.

1.5 Project Modules

Registration

Employee or Employer can register with valid details like contact details, experience details, profile details.

Search

Employee Can Search job according to their interest. And also apply for that job. Employer search candidates for their requirements using keyword. Employer also can communicate with employee for their any other query or information via send message.

Job Post

Employer post job for their organization. And include job vacancy, salary details, working hours, designation details, experienced details.

Manage Account

Employee can also delete his/her account anytime. Admin Can Manage Employee and Employer Details. Admin observed Users Action like job posting, candidate details false or not.

PROPOSED SYSTEM

Software design sits at the technical kernel of the software engineering process and is applied regardless of the development paradigm and area of application. Design is the first step in the development phase for any engineered product or system. The designer's goal is to produce a model or representation of an entity that will later be built. Beginning, once system requirements have been specified and analyzed, system design is the first of the three technical activities - design, code and test that is required to build and verify software.

The importance can be stated with a single word "Quality". Design is the place where quality is fostered in software development. Design provides us with representations of software that can assess for quality. Design is the only way that we can accurately translate a customer's view into a finished software product or system. Software design serves as a foundation for all the software engineering steps that follow. Without a strong design we risk building an unstable system – one that will be difficult to test, one whose quality cannot be assessed until the last stage. During design, progressive refinement of data structure, program structure, and procedural details are developed reviewed and documented. System design can be viewed from either technical or project management perspective. From the technical point of view, design is comprised of four activities – architectural design, data structure design, interface design and procedural design.

Job Procurement: Old and New Ways Job seeking usually involves different ways to look for jobs such as through personal contacts, direct telephone calls to employers, job agency office, scanning online job listings, etc. Before the Internet, became widely used as a method of seeking jobs, jobseekers spent a lot of time using various methods to look for job openings. Today, jobseekers use online methods which are very convenient and save a lot of time. Galanaki lists the following methods to be the traditional (old) ways for recruitment:

- 1) Employment recruitment agencies.
- 2) Job fairs.
- 3) Advertising in the mass media such as newspapers.
- 4) Management Consultants.
- 5) Advertisement in television and radio.
- 6) Existing employee contacts.
- 7) Schools colleges or universities students services department.
- 8) Workers or professional referrals.

These old job seeking methods are too slow, stressful, challenging and also lack quality. In addition, the applicants have to consider the cost and the amount of time to get the information they need, and other preparations they have to make. Finding all available job vacancies is a main step at in the job-seeking process. The Internet is now a powerful tool that jobseekers can use. Today, there are many sites that advertise job positions to be filled by people with certain skills in various fields. The Internet plays an important role in the area of human resource planning and development. Most planning and development organizations are now using computer technology and the Internet for staff recruitment. It should be noted that although the Internet has facilitated the process of job-seeking, it has not replaced the traditional methods, completely.

Importance of Job Portals

In the age of technology, the Internet has become the main source of information for jobseekers. Large corporations, Institutions, and universities include information on career Prospects on their websites. According to a survey, 70% of the workforce uses websites or portals on the Internet to Search for jobs in France. These websites or portals provide search engine to access information on job opportunities.

questions in order to arrive at the proper solution. Because the structures and responses are all pre-determined, you have complete control over the discourse.

2.1 Project Feasibility Study

Technical Feasibility

Technical feasibility study is concerned with specifying equipment and software that will successfully satisfy the user requirement; the technical needs of the system may vary considerably. The facility to produce outputs in a given time. Our project is a web based application which is based on client-server based application. In this application every page as output is render from server to client so it is necessary that the page should be rendered in time. For this I have avoided more and more code in the page- load event.

Economical Feasibility

Economical feasibility is the measure to determine the cost and benefit of the proposed system. A project is economical feasible which is under the estimated cost for its development. These benefits and costs may be tangible or intangible. Job Portal is the cost-effective project in which there is less possibility of intangible cost so there is no difficulty to determine the cost of the project.

Operational Feasibility

Operation feasibility is used to check whether the project is operationally feasible or not. Our project is mainly different from the other system because of its web-support feature. So the measure for operational feasibility is something different from other system. Generally the operational feasibility is related to organization aspects. The change determination is as such that early product were either a man or group of men or the jobs based manual but now a day with the advent of Internet technology.

2.2 Detailed Module Description with all Functionalities

Registration

In the registration module job seeker have to include all the details like personal details, contact details, education details like school, graduation, post-graduation, course certification details etc. Also job seeker has to add his experience details, job requirements and uploading resume and photo. While job recruiter has to add his contact details and organization details for the registration and upload company logo and profile.

Job Post

Employer can post a job by providing all the job details like qualifications details, requirements for the job, designation details, job salary details and also provide type of jobs. They also can delete the jobs whenever they want. After successfully posted a job it will be available for all the job seekers who are searching for a job. And it will be available on home page as recently posted job.

Search

Employee Can Search job according to their interest. And also apply for that job or they can add into wishlist for future whenever they find for job for that company then they easily find out company from wishlist. Employer search candidates for their requirements using keyword like technology. And also can communicate with employee for their any other query or information via send message .and also employer see the resume of applicants.

Manage Account

While employers can manage their job postings. And providing all the job details like qualifications details, requirements for the job, designation details, job salary details and also provide type of jobs. They also can delete the jobs whenever they want. While employee can manage their wishlist, applied for job and also getting full details of

employer. Employees can delete their account anytime. Also they can apply for the different jobs according to their interests

2.3 System Design

A data flow diagram is graphical tool used to describe and analyze movement of data through a system. These are the central tool and the basis from which the other components are developed. The transformation of data from input to output, through processed, may be described logically and independently of physical components associated with the system. These are known as the logical data flow diagrams. The physical data flow diagrams show the actual implements and movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams. Using two familiar notations Yourdon, Gane and Sarson notation develops the data flow diagrams. Each component in a DFD is labeled with a descriptive name. Process is further identified with a number that will be used for identification purpose. The development of DFD'S is done in several levels. Each process in lower level diagrams can be broken down into a more detailed DFD in the next level. The lop-level diagram is often called context diagram. It consists a single process bit, which plays vital role in studying the current system. The process in the context level diagram is exploded into other process at the first level DFD.

The idea behind the explosion of a process into more process is that understanding at one level of detail is exploded into greater detail at the next level. This is done until further explosion is necessary and an adequate amount of detail is described for analyst to understand the process. Larry Constantine first developed the DFD as a way of expressing system requirements in a graphical from, this lead to the modular design.

A DFD is also known as a “bubble Chart” has the purpose of clarifying system requirements and identifying major transformations that will become programs in system design. So it is the starting point of the design to the lowest level of detail. A DFD consists of a series of bubbles joined by data flows in the system.

2.4 DFD symbols

In the DFD, there are four symbols

- A square defines a source(originator) or destination of system data
- An arrow identifies data flow. It is the pipeline through which the information flows
- A circle or a bubble represents a process that transforms incoming data flow into outgoing data flows.
- An open rectangle is a data store, data at rest or a temporary repository of data

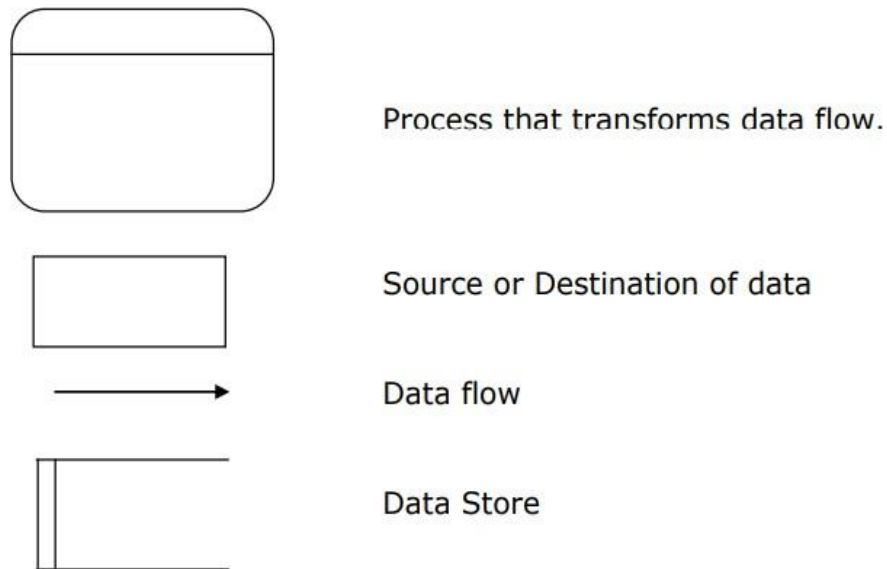


Fig 2.1. DFD Symbols

2.5 Data Flow Diagrams (DFDs)

A data flow diagram (DFD) is a graphical or visual representation using a standardized set of symbols and notations to describe a business's operations through data movement. They are often elements of a formal methodology such as Structured Systems Analysis and Design Method (SSADM). Superficially, DFDs can resemble flow charts or Unified Modeling Language (UML), but they are not meant to represent details of software logic.

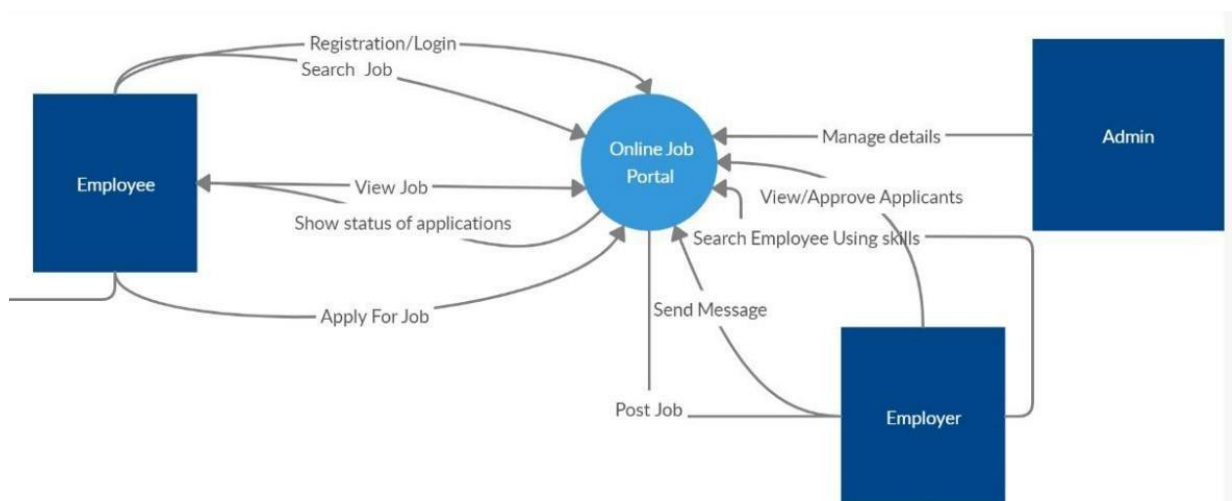


Fig 2.2. DFD level 0

This level shows the overall context of the system and its operating environment and shows the whole system as just one process.

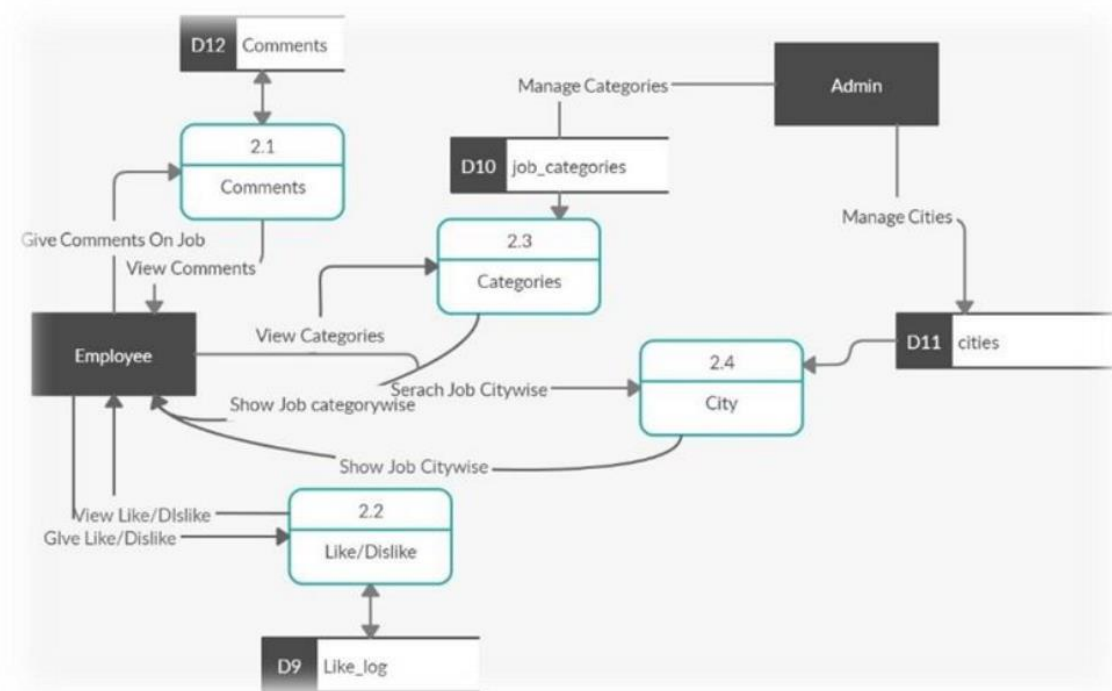


Fig 2.3. DFD level 1

Above is the DFD of the “Proposed System” we have refined the system up to two levels. Each break-up has been numbered as per the rule of DFD. We have tried to incorporate all the details of the system but there is some chance of further improvisation because of the study that is still going on for the project development.

TECHNOLOGIES USED

Job Searching web application is a professional job board that helps the employer, agent and job seeker to find their talents, dream jobs within a short time with its advanced search engine. It comes with modern design and clean code to help you extend it further. Job Portal project is web application built using PHP, MySQL as backend and HTML, CSS & JavaScript as technologies. Note that this software is developed as an academic project.

HTML

HTML stands for Hyper Text Markup Language

What is HTML?

- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

CSS

CSS stands for Cascading Style Sheets. CSS saves a lot of work. It can control the layout of multiple web pages all at once.

What is CSS?

Cascading Style Sheets (CSS) is used to format the layout of a webpage.

With CSS, you can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!

JAVASCRIPT

Javascript is used by programmers across the world to create dynamic and interactive web content like applications and browsers. JavaScript is so popular that it's the most used

programming language in the world, used as a client-side programming language by 97.0% of all websites

What is JAVASCRIPT?

- JavaScript is the **Programming Language** for the Web.
- JavaScript can update and change both **HTML** and **CSS**.
- JavaScript can **calculate**, **manipulate** and **validate** data.

PHP

PHP is a server scripting language, and a powerful tool for making dynamic and interactive Web pages. PHP is a widely-used, free, and efficient alternative to competitors such as Microsoft's ASP.

What is PHP?

- PHP is an acronym for "PHP: Hypertext Preprocessor"
- PHP is a widely-used, open source scripting language
- PHP scripts are executed on the server

- PHP is free to download and use

Node.js

Node.js is an open source server environment. Node.js allows you to run JavaScript on the server.

What is Node.js?

- Node.js is an open source server environment
- Node.js is free
- Node.js runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.)
- Node.js uses JavaScript on the server

USE CASE DIAGRAM

4.1 Use Case Diagram

UML is an acronym that stands for Unified Modeling Language. Simply put, UML is a modern approach to modeling and documenting software. In fact, it's one of the most popular business process modeling techniques. In the Unified Modeling Language (UML), a use case diagram can summarize the details of your system's users (also known as actors) and their interactions with the system. To build one, you will use a set of specialized symbols and connectors.

A diagram is a visualization of set of elements and the relationships between them. Use case is a set of scenarios, which defines functionalities of the system from a user's perspective. The main components of a use case diagram include actors, use cases and their relationships. They depict the interaction between actors and system to achieve certain goal. This, a use case diagram is important in modelling the behavior of a system.

Following are the purposes of a use case diagram given below:

- It gathers the system's needs.
- It depicts the external view of the system.
- It recognizes the internal as well as external factors that influence the system.
- It represents the interaction between the actors.

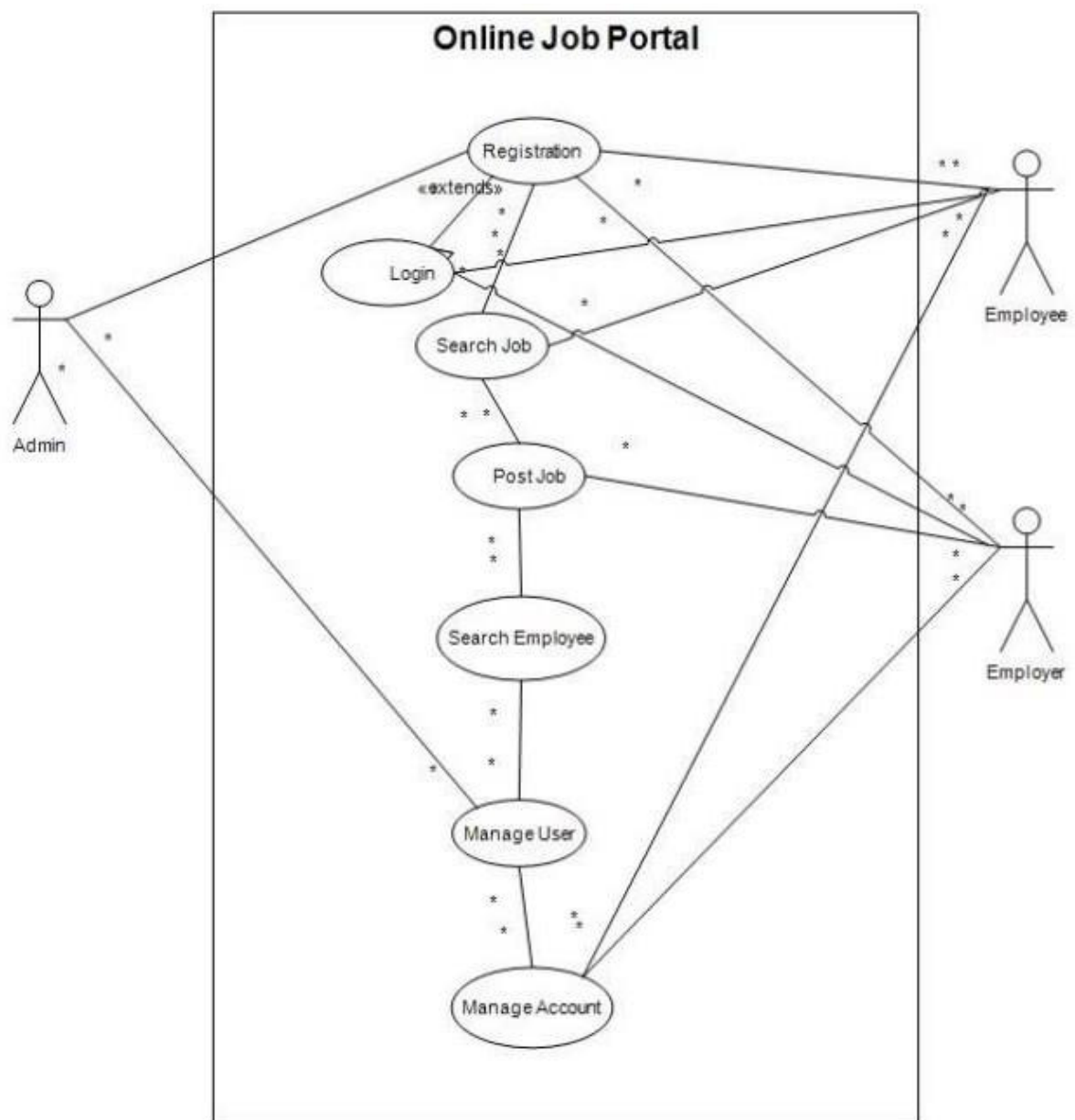


Fig 3.3. Use Case Diagram

SYSTEM IMPLEMENTATION DETAILS

5.1 Implementation

The Online Job Search Portal is a web-based and android application, which revolutionizes the way companies hire the candidates and jobseekers search for job vacancies. The employers can view reviews given by the jobseekers and make improvements in their system accordingly. The application provides a flexible and easy to use environment on desktops as well as portable devices like smart phones/tablets for the users to achieve their respective objective. The modules that I have implemented in the Job Search Portal are as listed.

- Employer
- Jobseeker

Employer

Employer users will be able to perform functions such as registering with the application and creating an account by providing the details of Employer Name, Employer Code, Address, Company E-mail, Mobile Number, Login Name, and Password that are stored in the Employer table of MySQL database. Once the account is activated, this module allows employers to post jobs summarizing responsibilities and expected skills that will be saved in the Posted_Jobs table of MySQL database. The employer will also be given privilege to activate or deactivate jobs. He/she can view the list of job postings that are active. He/she can also view the applicant details that have applied for a particular job posting. The employer will be able to view reviews provided by the jobseeker.

Jobseeker

The Jobseeker users will be able to perform functions such as registering with the application and creating an account by providing the details of First Name, Last Name, E-mail, Password, Mobile Number, Primary Skill and Experience that are stored in the Jobseeker table of MySQL database. 19 Once the account is activated, jobseekers can search, view and apply for active job openings. All the applied jobs details are stored in the Applied_Jobs table of MySQL database. The applicants can also write/update reviews for the companies. The Online Job Search web application is developed on Eclipse IDE using Java 1.8 and Spring MVC framework. I have used annotation configuration for spring replacing XML configurations. The application uses Hibernate framework for mapping object-oriented domain model to a relational database. The backend database for the web application is MySQL and I have used MySQL workbench and MySQL query browser for managing the database. The web application is deployed in the tomcat server. An android application is developed for the jobseeker module using Android Studio IDE, Android SDK framework and Java using the RESTful web services provided by the web application.

5.2 REQUIREMENTS AND SPECIFICATION

Hardware Specification

System Configuration: Windows 11, 64-bit operating system, x64-based processor, Processor: Intel core i5, Processor speed: 3.40 GHz, RAM: 8 GB

Software Specification

Below are the technical requirements to develop Job Search Portal application

- Operating System: Windows 11
- IDE: Visual Studio IDE
- Database: MySQL
- Front End: HTML5, CSS3, Bootstrap, JavaScript, jQuery
- Back End: NodeJs, PHP, Django
- Browser: Google Chrome or Firefox

CONCLUSION

It has been a great pleasure for us to work on this exciting and challenging project. This project proved good for us as it provided practical knowledge of not only programming, web based application and to some extent Windows Application and SQL Server, but also about all handling procedure related with “JOB SEARCHING WEB”. It also provides knowledge about the latest technology used in developing web enabled application and client server technology that will be great demand in future. This will provide better opportunities and guidance in future in developing projects independently.

Job Search Portals stands as a revolutionizing element in the sphere of recruitment. They act as a communication bridge between applicants and recruiters facilitating their requirements. This application helps organizations to have a greater exposure to the candidate pool and also job seekers facilitating wide search of jobs matching their interests. The android application provides flexibility to the jobseekers to view the openings and applied jobs without the need to carry a laptop. This application provides an enhanced user experience for both employer and jobseeker. It provides user friendly interface which facilitates in reaching wide range of audience. The application has achieved all the requirements that were initially set in the requirements gathering phase. This project taught me some best practices in the technology stack like Spring MVC, Hibernate ORM, Android development, RESTful web services. Starting from requirements elicitation to design, construction, implementation and testing, I have gained a very good experience working with various technologies at every phase. Development of this project boosted my confidence in mobile and web development.

BENEFITS

The project is identified by the merits of the system offered to the user. The merits of this project are as follows

- It is a web-enabled project.
- This project offers user to enter the data through simple and interactive forms. This is very helpful for the client to enter the desired information through so much simplicity.
- The user is mainly more concerned about the validity of the data, whatever he is entering. There are checks on every stages of any new creation, data entry or updation so that the user cannot enter the invalid data, which can create problems at later date.
- Sometimes the user finds in the later stages of using project that he needs to update some of the information that he entered earlier. There are options for him by which he can update the records. Moreover there is restriction for his - 47 - that he cannot change the primary data field. This keeps the validity of the data to longer extent.

- User is provided the option of monitoring the records he entered earlier. He can see the desired records with the variety of options provided by him.
- From every part of the project the user is provided with the links through framing so that he can go from one option of the project to other as per the requirement. This is bound to be simple and very friendly as per the user is concerned. That is, we can say that the project is user friendly which is one of the primary concerns of any good project.
- Data storage and retrieval will become faster and easier to maintain because data is stored in a systematic manner and in a single database.
- Decision making process would be greatly enhanced because of faster processing of information since data collection from information available on computer takes much less time than manual system.
- Allocating of sample results becomes much faster because at a time the user can see the records of last years.
- Easier and faster data transfer through latest technology associated with the computer and communication.
- Through these features it will increase the efficiency, accuracy and transparency.

LIMITATIONS

- The size of the database increases day-by-day, increasing the load on the database back up and data maintenance activity.
- Training for simple computer operations is necessary for the users working on the system.

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